Q.25 Explain any two.

a) Heat loss from a pipe

c) Fouriers's law

d) Feeding arrangement in MEE

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Roll No. ....

4th Semester/ Chemical (Pulp & Paper)
Subject: Heat Transfer

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Unit of Heat Flow rate is

a) Watt/Square meter b) Watt

c) Watt/Cubic meter d) All of these

Q.2 What is unit of q in this equation $q = -kAT/x$?

a) Watt/square meter b) Watt/cubic meter

c) Watt d) All of these

Q.3 Materials having very low thermal conductivity are called_______.

a) Fins b) Insulators

c) Conductors d) None

Q.4 What are the unit of temperature gradient

a) Meter b) Litre

c) Centigrade/Metre d) None

Q.5 Fouriers's law is applicable in which mode of heat transfer

a) Conduction b) Radiation

c) Convection d) None

Q.6 Write the full form of LMTD

a) Long mean temp. data

b) Long mean temp. Detail

c) Long mean temp. Difference

d) None

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write one example of conduction.

Q.8 Heat transfer due to bulk motion of fluid is called______.

Q.9 What is the formula of Reynolds number?

Q.10 Write the driving force in conduction heat transfer.

Q.11 The emissive power of black body is given by _______.

Q.12 What is the flow of fluids in counter current heat exchanger.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Define unsteady state heat conduction.

Q.14 State and explain weins displacement law of heat radiation.

Q.15 Write a short note on dimensional analysis.

Q.16 Define nusselt number & Stanton Number.

Q.17 Explain about the critical thickness of insulation.

Q.18 What is reflectivity. Explain in brief.

Q.19 Write any five properties required for good conducting material.

Q.20 Draw neat sketch of double pipe heat exchanger.

Q.21 Write a short note solar radiation?

Q.22 Define insulation. Name common insulators.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Derive an expression of heat transfer by conduction through a hollow cylinder when thermal conductivity is constant.

Q.24 State and explain kirchoff's law of radiation with the help of neat diagram.